

Excluding the Bounded Negative Part

Lean-checked rigidity for Erdős Problem #243

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ABSTRACT

Let $a_1 < a_2 < \dots$ be positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and rational reciprocal sum, and put $P_n = \prod_{j < n} a_j$. We prove that

$$\limsup_{n \rightarrow \infty} \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < +\infty$$

forces $a_{n+1} = a_n^2 - a_n + 1$ eventually. In the integral tail state, bounded negative error caps upward increments, while normalised vanishing forces a nonterminal numerator to infinity. Stabilising the common divisor then supplies a Chinese remainder block that those increments cannot cross. An LCM formulation gives a weighted record criterion and the corresponding LCM-prefactor consequence. The required bound or record budget remains unproved for the unrestricted problem.

1 Introduction

Corollary 1.1 (original-coordinate bounded defect). *Let $a_1 < a_2 < \dots$ be positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j < n} a_j$. If*

$$\limsup_{n \rightarrow \infty} \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < +\infty,$$

then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .

The statement is [Lean source](#).

The obstruction. The equation $C_{n+1} = C_n - E_n$ translates a bound on negative error into a bound on upward increments. At negative indices the common divisor of C_n, D_n divides the bounded magnitude. Once this divisor stabilises, reduction makes the later numerators coprime to the old multipliers. The Chinese remainder theorem places consecutive forbidden heights beyond any fixed prefix. A first crossing then gives the contradiction. Section 6 uses the same divisibility at record sources, so that only record excess is charged.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Write $\text{ctr}(a, D, C) = D - (a - 1)C$. The following integer-state theorem implies Corollary 1.1; the ordinary tail transfer is proved in Section 4. Its centring hypothesis is redundant under normalised vanishing and is retained to match the source statement.

Theorem 1.2 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ and $E : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy*

1. $a_n > 1$ and $C_n > 0$ for every n ;
2. the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;
3. $E_n = \text{ctr}(a_n, D_n, C_n)$ for every n ;
4. eventual strict centring: $|E_n| < C_n$ for all large n ;
5. eventually bounded negative part: $-B \leq E_n$ for all large n , for some B ;
6. normalised vanishing: for every K there is an N with $K|E_n| < C_n$ for all $n \geq N$.

Then $E_n = 0$ for all sufficiently large n .

Problem 1.3 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

The additional hypothesis in Theorem 1.2 is the lower bound on E_n . The growth and rationality assumptions already supply the other state hypotheses after a finite shift. Koizumi's nonnegative-error case is integer descent [8, Proposition 1(2)]; the theorem allows arbitrary sign changes with bounded negative depth. Bado's author-posted preprint assumes two-sided bounded errors [7, Theorem 5.1]. The comparisons concern these stated hypotheses and make no priority claim.

2 Proof of bounded-negative rigidity

The arithmetic ingredients are proved immediately afterwards. Normalised vanishing and integrality give $C_n \rightarrow \infty$ on a nonzero tail, while the lower error bound gives $C_{n+1} \leq C_n + B$.

Proof of Theorem 1.2. Suppose not. We first remove every late zero: by Theorem 3.2 and its contrapositive form, E is nowhere zero beyond the centring threshold. Two cases remain. If E is negative cofinally, then (5) bounds those magnitudes, so Proposition 3.7 makes the tail gcd stabilise and the orbit may be taken reduced past that point. Hypothesis (5) with Proposition 3.1 gives $C_{n+1} \leq C_n + B$, a bounded rise; hypothesis (6) with E nowhere zero gives $C_n \rightarrow \infty$. A reduced exact tail with a divergent numerator and bounded rise is excluded by Theorem 3.5. It remains to treat the case that E is eventually positive, and there the descent of Theorem 3.2 applies and forces E to vanish, contradicting nowhere-vanishing. $\square$

Corollary 2.1. *Under Theorem 1.2, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ eventually.*

Proof. Apply the two-zero implication in Corollary 3.3 below. $\square$

3 The arithmetic ingredients

3.1 Error, absorption and descent

Write $\Delta_n = a_{n+1} - (a_n^2 - a_n + 1)$.

Proposition 3.1 (error identities). *For an exact integer state,*

$$C_{n+1} = C_n - E_n, \quad \Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}.$$

Proof. Substitute $E_n = D_n - (a_n - 1)C_n$ and use $D_{n+1} = a_n D_n$, $C_{n+1} = a_n C_n - D_n$. $\square$

Both identities are [Lean source](#).

Theorem 3.2 (absorption and descent). *For a positive exact state with strict centring, $E_n = 0$ implies $E_{n+1} = 0$. For any positive integer state with $C_{n+1} = C_n - E_n$, eventual nonnegativity of E_n implies its eventual vanishing.*

Proof. When $E_n = 0$, the second identity makes E_{n+1} a multiple of C_{n+1} ; strict centring forces that multiple to be zero. In the second assertion, C_n is eventually a nonincreasing sequence of positive integers, so it stabilises. $\square$

Both assertions are [Lean source](#).

Corollary 3.3 (two zero errors). *If $E_n = E_{n+1} = 0$ and $C_{n+1} \neq 0$, then $a_{n+1} = a_n^2 - a_n + 1$. Thus eventual zero error in a positive exact state implies the eventual Sylvester recurrence.*

Proof. The second error identity has nonzero factor C_{n+1} . $\square$

The local and eventual forms are [Lean source](#).

Absorption holds after any eventual centring threshold. On a nonterminal tail it removes all subsequent zeros. Then $|E_n| \geq 1$, so normalised vanishing gives $C_n \rightarrow \infty$.

3.2 The first-crossing obstruction

Lemma 3.4 (consecutive multiples). *For pairwise coprime integers $m_0, \dots, m_{B-1} \geq 2$ and every lower bound, there is a larger t such that $m_i \mid t + i$ for each $i < B$.*

Proof. Solve $t \equiv -i \pmod{m_i}$ by the Chinese remainder theorem, then add multiples of $\prod_i m_i$. $\square$

Theorem 3.5 (bounded-rise obstruction). *Let $u_n \in \mathbb{N}$ tend to infinity with $u_{n+1} \leq u_n + B$, for a fixed integer $B > 0$. There is no infinite pairwise coprime family $m_i \geq 2$ with $\gcd(m_i, u_t) = 1$ whenever $i < t$.*

Proof. Choose B old moduli and take a CRT block $[t, t + B)$ beyond every numerator in the prefix before they become old. At the first crossing of t , the new numerator lies in that block, so it is divisible by an old modulus. This contradicts coprimality. No monotonicity of u is used. $\square$

3.3 Reduction and stabilisation

A reduced exact tail has positive u_n , integer $v_n \geq 0$, $\gcd(u_n, v_n) = 1$, and

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n.$$

Proposition 3.6 (persistent coprimality). *In a reduced exact tail, $\gcd(a_n, v_n) = 1$. Distinct multipliers are pairwise coprime, and every earlier multiplier is coprime to every later numerator.*

Proof. A common prime divisor of a_n, v_n would divide both u_{n+1} and v_{n+1} . Also, $a_i \mid v_t$ for $i < t$, so reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. $\square$

All three conclusions are [Lean source](#).

Proposition 3.7 (gcd stabilisation). *For a positive exact state, if negative errors have bounded magnitudes along a cofinal set of indices, then $G_n = \gcd(C_n, D_n)$ eventually stabilises. Division by its stable value gives a reduced exact tail.*

Proof. Both updates preserve common divisors, so $G_n \mid G_{n+1}$. Moreover $G_n \mid E_n$, hence $G_n \leq -E_n$ at a negative index. The positive divisibility chain is bounded along a cofinal set and therefore bounded everywhere; it eventually stabilises. Division by the stable value preserves both exact updates and leaves the states coprime. $\square$

The stable gcd and the reduced exact tail are [Lean source](#).

In the main proof, division by the stable gcd preserves divergence and a bounded upward increment. Theorem 3.5 therefore applies to the reduced numerator. Sparse changes of the gcd without eventual stabilisation are a different statement; the supporting record retains that distinction and its finite-block consequences.

4 Transfer to a reciprocal series

Let $x_n = \sum_{k \geq n} 1/a_k$, suppose $x_1 = p/q$ with $q > 0$, and put

$$P_n = \prod_{j < n} a_j, \quad D_n = qP_n, \quad C_n = D_n x_n.$$

Each C_n is a positive integer, since D_n clears the rational sum and every term of the preceding finite prefix. The tail relation $x_n = 1/a_n + x_{n+1}$ gives the exact state updates.

Quadratic growth gives $a_n \geq \exp(c2^n)$ eventually for some $c > 0$ and

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}).$$

Consequently

$$\frac{C_{n+1}}{C_n} = \frac{a_n x_{n+1}}{x_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n) \rightarrow 1, \quad \frac{E_n}{C_n} \rightarrow 0.$$

Thus strict centring also holds eventually. These are the ordinary canonical-tail implications of Koizumi's Corollary 3 and Lemma 4 [8].

Proof of Corollary 1.1. Put $\gamma_n = a_n^2/a_{n+1} - 1$. Since

$$\frac{P_{n+1}/a_{n+1}}{P_n/a_n} = 1 + \gamma_n \rightarrow 1, \quad P_n/a_n = \exp(o(n)),$$

the two-term estimate gives

$$E_n + q \frac{P_n}{a_n} \gamma_n = \frac{qP_n}{a_{n+1}} + O\left(\frac{qP_n a_n}{a_{n+1}^2}\right) = o(1). \quad (1)$$

The assumed upper bound therefore gives an eventual lower bound on the integer E_n . After deleting a finite prefix, all hypotheses of Theorem 1.2 hold. Apply Corollary 2.1. $\square$

The same calculation is valid with qP_n replaced by any positive clearance $H_n \leq qP_n$: if $W_n = H_n x_n$ and $Z_n = H_n - (a_n - 1)W_n$, then

$$Z_n + \frac{H_n}{a_n} \gamma_n = o(1). \quad (2)$$

This will supply the LCM-prefactor consequence below.

5 A scalar finite-mass criterion

Theorem 5.1 (finite negative relative mass). *Let C_n be positive integers and E_n integers satisfying $C_{n+1} = C_n - E_n$. If*

$$\sum_n \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. No denominator dynamics or normalised vanishing is required.

Proof. Set $\delta_n = (-E_n)_+/C_n$. Then

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer upper bound K for C_n . Each strict rise contributes at least $1/K$ to $\sum \delta_n$, so there are only finitely many rises. The remaining positive integer sequence is nonincreasing and stabilises. $\square$

On an exact reciprocal-tail orbit the conclusion gives the Sylvester recurrence. The scalar criterion and bounded negative error are distinct quantities to estimate; on the common exact-orbit class with normalised vanishing they are each equivalent to the same endpoint.

6 Weighted first crossings of LCM records

Only steps setting an LCM numerator record need contribute. The first-crossing argument permits subtraction of a fixed baseline and any decreasing weight with divergent integral. The theorem in this section has an ordinary proof.

Set

$$L_n = \text{lcm}(q, a_0, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The rational tail has denominator dividing L_n , so U_n and V_n are integers. With $\rho_n = \gcd(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Centring implies that every sufficiently late strict rise has $\rho_n = 1$: otherwise $U_{n+1} \leq 3U_n/4$. Its actual jump is therefore $d_n = U_{n+1} - U_n = -V_n$.

Theorem 6.1 (weighted record excess). *Assume the growth and rationality hypotheses of Problem 1.3. Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Proof. Suppose the sequence is not eventually Sylvester. Absorption and integrality give $1/U_n \leq |V_n|/U_n = |E_n|/C_n \rightarrow 0$, so there are infinitely many record steps. Every sufficiently late record is fresh. Their digits are pairwise coprime: an earlier digit divides the later L_n , while the digit at a fresh record is coprime to L_n . For any fixed $B \geq 1$, choose B such digits $m_0, \dots, m_{B-1} > B$ and take T after their indices.

Put $P = \prod_i m_i$ and choose x by the Chinese remainder theorem with $m_i \mid x + i$. Consider all translates $\tau = x + B + kP > R_T$, $k \in \mathbb{Z}$; the first is at most $R_T + P$. A first crossing $U_n \leq R_n < \tau \leq U_n + d_n$ is a record step. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so some m_i divides U_n . It also divides L_n , hence divides $d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

If the step first crosses $h \geq 1$ such heights, their spacing gives $(h - 1)P < d_n$. With $r = d_n - B \geq 1$ and $P \geq B + 1$, we have $d_n = B + r \leq Pr$, whence $h \leq r$. Monotonicity of f now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each height above R_T has one first crossing. Summing, and comparing each interval of length P with its left endpoint, yields the finite bound

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T + P}^{R_N} f(t) dt. \quad (3)$$

Since $R_n \rightarrow \infty$, the right side diverges for every $B \geq 1$; $B = 0$ follows by domination. Conversely a Sylvester tail telescopes to $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (3) is $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n$. This is summable because $U_n \leq C_n = \exp(o(n))$, $f(U_n) \leq f(1)$, and $a_n \geq \exp(c2^n)$ eventually for some $c > 0$. Consequently Theorem 6.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (4)$$

for some B . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. Also, d_n is the actual jump, including any recovery from a drawdown; it must not be replaced by $R_{n+1} - R_n$ in the crossing proof.

Corollary 6.2 (LCM-weighted bounded defect). *Assume the hypotheses of Problem 1.3. Write $A_n = \text{lcm}(a_1, \dots, a_{n-1})$ with $A_1 = 1$ and*

$$Z_n = \frac{A_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right).$$

If $\limsup Z_n < \infty$, then the sequence is eventually Sylvester.

Proof. Put $t_n = L_n/A_n = q/\gcd(q, A_n)$, so $1 \leq t_n \leq q$. Equation (2), with $H_n = L_n$, gives $V_n + t_n Z_n = o(1)$. A finite upper bound on Z_n therefore bounds the negative part of V_n . The record series vanishes after a finite prefix for a sufficiently large baseline, so Theorem 6.1 applies. $\square$

Erdős–Straus Theorem 3 uses a nonpositive limsup in the corresponding next-index LCM expression [1]. Here any finite upper bound suffices under the quadratic-limit assumption. Since $A_n \mid P_n$, this also implies Corollary 1.1: multiplication by a factor in $(0, 1]$ preserves an upper bound, including at negative values.

The divisor constraint also survives integral numerator coefficients: $d = (a_n - 1)U_n - b_n L_n$ is divisible by every divisor of both U_n, L_n . The already-proved coefficient-uniform variant gives bounded LCM height from a bounded negative part; normalised vanishing is used afterwards for stationarity. Its proof and exact bounded-height counterexamples are in the [coefficient proof supplement](#).

7 Further consequences

Theorem 7.1 (cubic-rate irrationality). *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

Proof by the polynomial exclusion. Under rationality, the canonical estimate gives $C_{n+1}/C_n = 1 + 3/n + o(n^{-3})$. Integer finite differences force $C_n = An(n+1)(n+2) + B$ eventually. The fixed-cubic exclusion rules this out. The complete ordinary polynomial and number-field argument is retained in [the proof supplement, Section 8](#); finite congruence checks alone are insufficient for that implication. $\square$

The same finite-difference extraction excludes every nonintegral $\lambda > 1$ under the rate $a_n^2/a_{n+1} = 1 + \lambda/n + o(n^{-\lambda})$: eventual polynomial growth would force its degree to equal λ . These rates lie outside Koizumi's $1 + o(1/n)$ hypothesis [8, Remark 3].

There is also an admitted quantitative extension. Under the hypotheses of Theorem 1.2 other than (5), put $\text{LL}(x) = \log_2 \log_2 \max(4, x)$. Each fixed $\delta \in (0, 1)$ gives

$$(-E_n)_+ \leq (1 - \delta) \text{LL}(C_n) \text{ eventually} \implies E_n = 0 \text{ eventually.}$$

The LCM version needs the corresponding bound only at late record steps. The proofs use a CRT block in $[P, 2P)$ and the canonical multiplier scale; they are retained in [the slow-negative proof](#) and [the record-only extension](#). These are ordinary results with stated rate hypotheses; neither establishes the general record budget.

8 The remaining arithmetic estimate

Proposition 8.1 (necessary profile). *The canonical state of a counterexample has $E_n \neq 0$ eventually, $|E_n|/C_n \rightarrow 0$, unbounded negative magnitudes along negative indices, and*

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Proof. Absorption excludes late zeros, descent excludes an eventually nonnegative error, and Theorems 1.2 and 5.1 exclude the two finiteness conditions. $\square$

The canonical necessary profile is [Lean source](#).

For $B \geq 0$ put

$$F_B(X) = \sum_{\substack{n \in \mathcal{R} \\ U_n \leq X}} (-V_n - B)_+.$$

The remaining equivalent estimate is

$$\exists B \in \mathbb{N} : \quad \liminf_{X \rightarrow \infty} \frac{F_B(X)}{X} = 0. \quad (5)$$

On a nonterminal orbit first crossings instead give $F_B(X) \geq X/P_B - O_B(1)$, with an orbit-dependent CRT modulus P_B . The missing step is to derive (5) from the canonical

dynamics. The equivalence with the existence of an admissible summable weight is proved in the [record-budget supplement](#); it does not establish the estimate.

The obstruction is the exact unit-numerator feedback at the record sources. The scalar sequence $C_n = n^2 + 1$, $E_n = -(2n + 1)$ has normalised vanishing, subexponential height and finite square mass, but is not an exact reciprocal-tail orbit: its numerator word $0, 0, 2 \pmod{5}$ violates the persistent-zero transport of the exact equations. Likewise, small rises and avoidance of an arbitrary sparse coprime family do not control the canonical timing and size of the available divisors. The supporting record keeps those falsifying examples with the exact hypotheses each preserves.

Formal scope and supporting record. The bounded-negative state theorem is checked in Lean 4 against Mathlib at the stated source checkpoint: [eventual bounded-negative rigidity](#). The canonical analytic transfer and the global weighted, polynomial and slow-negative arguments are ordinary proofs; the scalar finite-mass proof is printed in full above. The auxiliary source guide, constant and periodic exclusions, primitive feedback, finite certificates and unsuccessful global extensions belong to the accompanying reasoning record. Their stronger global producer hypotheses remain explicit.

Declarations. The authorship and AI-use statement on the first page applies to this manuscript. Code and mathematical source are available at the repository checkpoint `f36a98bf3d3e`. A source declaration establishes only its own statement; the ordinary transfer and the unresolved estimate above are separately identified.

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